

Enterprise Multi-Branch Network Project

1. Project Overview

Designed and implemented an enterprise-style multi-branch company network using Cisco Packet Tracer. The network includes HQ/Core infrastructure, branch offices, VLAN segmentation, dynamic routing, WAN connectivity, and enterprise security features.

2. Network Architecture

The network follows a Hub-and-Spoke WAN architecture connecting three branches to a central HQ using routers, switches, servers, firewall, and cloud/ISP simulation.

3. VLAN Design

VLANs were implemented to logically separate departments, reduce broadcast traffic, and improve security and network management.

4. IP Addressing Scheme

A structured IP addressing plan was implemented using dedicated branch networks and subnetted VLANs for scalability and organization.

5. Inter-VLAN Routing

Router-on-a-Stick was implemented using subinterfaces and 802.1Q encapsulation to enable communication between VLANs.

6. Dynamic Routing

OSPF dynamic routing was configured to allow automatic route exchange between branches and HQ for scalable enterprise routing.

7. WAN Connectivity

Serial WAN links connected all branches to HQ using /30 point-to-point subnets for efficient IP utilization.

8. DHCP Configuration

DHCP was configured to automatically assign IP addresses, default gateways, subnet masks, and DNS information.

9. Security Features

Security mechanisms included SSH secure management, ACL traffic filtering, Port Security on access ports, and firewall implementation for network protection.

10. Cloud Integration

A cloud device was used to simulate ISP and Internet connectivity in the enterprise environment.

11. Testing & Verification

Connectivity and functionality were verified using ping tests, OSPF neighbor verification, VLAN trunk verification, and routing table inspection.

12. Conclusion

The project successfully implemented a scalable enterprise network with VLAN segmentation, OSPF routing, WAN connectivity, centralized architecture, and layered security mechanisms.

VLAN Table

VLAN ID	Department
10	HR
20	Sales
30	IT
40	Servers
99	Management

Branch Addressing

Branch	Network
Branch 1	10.1.0.0/16
Branch 2	10.2.0.0/16
Branch 3	10.3.0.0/16